Homework 2

● Graded

Student

Colin Skinner

Total Points

99 / 100 pts

Question 1

Problem 1

19 / 20 pts

✔ **+ 10 pts** Correct Velocity

+ 10 pts Correct Acceleration

– 3 pts Minor calculation error

✔ **– 1 pt** Minor error in final answer

+ 5 pts Some correct elements in approach to velocity calculation

+ 5 pts Some correct elements in approach to acceleration calculation

💬 **+ 10 pts** Point adjustment

Question 2

Problem 2

20 / 20 pts

✔ **– 0 pts** Correct

– 3 pts Minor calculation error

– 5 pts Setup Error

– 17 pts Attempts at solution

– 0 pts See point adjustments

Question 3

Problem 3

20 / 20 pts

✔ **– 0 pts** Correct

– 3 pts Minor Calculation / Sign Error

– 7 pts Major Formulation Error

– 0.5 pts Did not write acceleration as a function of time

– 20 pts no attempt

Question 4

Problem 4

10 / 10 pts

✔ **– 0 pts** Correct

– 2.5 pts Leaves correct solution in cylindrical coordinates (not a sufficient solution for direction)

– 10 pts Incorrect, does not identify ODE

+ 1.5 pts Attempt at solution, although with incorrect approach/assumptions

– 2 pts Error in velocity magnitude

– 2 pts Error in direction

– 1 pt Small error

Question 5

Problem 5

15 / 15 pts

✔ **– 0 pts** Correct

– 4 pts Doesn't split moment between two pulleys

– 4 pts Misunderstanding of mass rate

– 4 pts Calculation error or error with assumed relations

– 2 pts Small calculation error

Question 6

Problem 6

15 / 15 pts

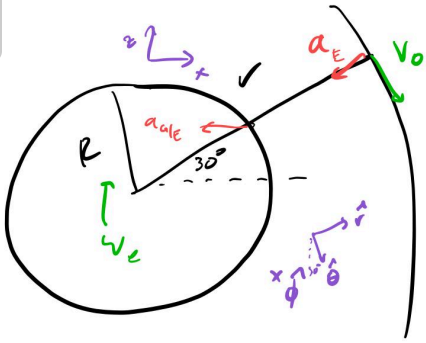
✔ **– 0 pts** Correct

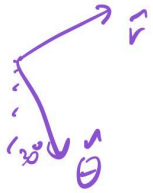
1. Greenwood 2-6

Question assigned to the following page: 1

ω_e rad/sec. As the satellite moves toward the equator, it passes directly over a radar station at 30° north latitude. This station measures the satellite's relative velocity and acceleration. Find this relative velocity and acceleration, expressing the results in terms of $\mathbf{e}_r$, $\mathbf{e}_\theta$, $\mathbf{e}_\phi$, where $\mathbf{e}_\theta$ points due south and $\mathbf{e}_\phi$ due east.

Hint: Make sure to find the velocity and acceleration of the satellite in the reference frame of the ground station that rotates with the Earth, and not the inertial reference frame.





Frames
G: ground
I: Earth inertial
S: satellite

$$\hat{z} = \sin 30^\circ \hat{r} - \cos 30^\circ \hat{\theta} = \frac{1}{2}\hat{r} - \frac{\sqrt{3}}{2}\hat{\theta}$$

$$\vec{v}|_S = V_0 \hat{\theta}$$

$$\vec{r} = r\hat{r} \qquad \vec{v}|_S = \dot{r}\hat{r} + r\dot{\hat{r}} = r(\vec{\omega} \times \hat{r}) = \omega_{sat} r \hat{\theta} \rightarrow \omega_{sat} = \frac{V_0}{r}\hat{\phi}$$

$$\vec{\omega}_{S/G} = \omega_e(-\hat{z}) = \omega_e\left(-\frac{1}{2}\hat{r} + \frac{\sqrt{3}}{2}\hat{\theta}\right) = -\frac{\omega_e}{2}\hat{r} + \frac{\sqrt{3}\omega_e}{2}\hat{\theta}$$

$$\vec{\omega}_{S/G} \times \vec{v} = \omega_e \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 0 \end{vmatrix} = \omega_e\left(-r\frac{\sqrt{3}}{2}\right)\hat{\phi} = -\frac{\sqrt{3}\omega_e r}{2}\hat{\phi}$$

$$\vec{v}_G = \vec{v}_S + \vec{\omega}_{G/S} \times \vec{r} \qquad \boxed{\vec{v}_G = V_0\hat{\theta} - \frac{\sqrt{3}\omega_e r}{2}\hat{\phi}}$$

$$\vec{a}_S = \frac{d}{dt}(\vec{v}_S) = V_0\dot{\hat{\theta}} = V_0(\vec{\omega}_{sat} \times \hat{\theta}) = V_0(\omega_{sat}\hat{\phi} \times \hat{\theta})$$

$$= V_0\omega_{sat}(-\hat{r}) = V_0\left(\frac{V_0}{r}\right)(-\hat{r}) = -\frac{V_0^2}{r}\hat{r}$$

$$\vec{\omega}_{S/G} \times \vec{v}_G = \omega_e \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & V_0 & -\frac{\sqrt{3}\omega_e r}{2} \end{vmatrix} = -\frac{3\omega_e^2 r}{4}\hat{r} - \frac{\sqrt{3}\omega_e^2 r}{4}\hat{\theta} - \frac{V_0\omega_e}{2}\hat{\phi}$$

$$\vec{a}_G = \vec{a}_S + \vec{\omega}_{S/G} \times \vec{v}_G$$

$$\boxed{\vec{a}_G = \left(-\frac{V_0^2}{r} - \frac{3\omega_e^2 r}{4}\right)\hat{r} - \frac{\sqrt{3}\omega_e^2 r}{4}\hat{\theta} - \frac{V_0\omega_e}{2}\hat{\phi}}$$

Question assigned to the following page: 2

inward from the vertical. Find the acceleration of a tack in the tire (0.4 m radius) as it passes through the highest point of its path. Use cylindrical unit vectors in expressing the answer.

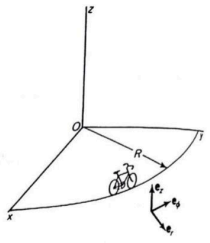
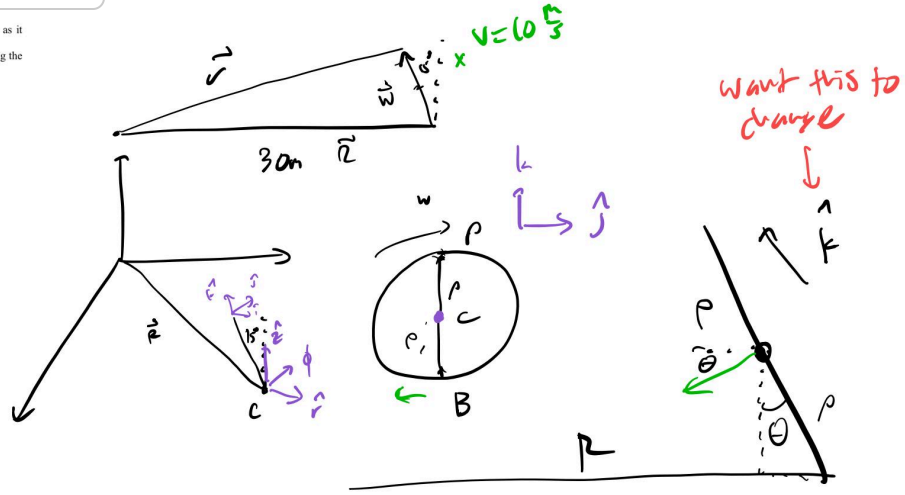


Figure P2-8



C: center P: pin
O: origin

$$\vec{r}_{B/O} = R\hat{r} \qquad \vec{v}_{B/O} = R\Omega\hat{z}\times\hat{r} = R\Omega\hat{\phi} = v\hat{\phi} \qquad \Omega = \frac{v}{R}$$

$$\vec{\Omega} = \Omega\hat{z} \qquad \vec{\omega} = -\omega\hat{i} \qquad \vec{\omega}_{tot} = \vec{\Omega} + \vec{\omega} = \Omega\hat{z} - \omega\hat{i}$$

$$\vec{r}_{C/O} = R\hat{r} - \rho\sin\theta\hat{r} + \rho\cos\theta\hat{z} - \rho\hat{k}$$

$$\vec{v}_{C/O} = R\dot{\hat{r}} - \rho\sin\theta\dot{\hat{r}} + \rho\cos\theta\dot{\hat{z}} - \rho\left[(\Omega\hat{z} - \omega\hat{i})\times\hat{k}\right]$$

$$\vec{v}_{C/O} = R\Omega\hat{\phi} - \rho\Omega\sin\theta\,\hat{\phi} - \rho\Omega\hat{z}\times\hat{k} + \rho\omega\hat{i}\times\hat{k}$$

$$= R\Omega\hat{\phi} - \rho\Omega\sin\theta\,\hat{\phi} - \rho\Omega\sin\theta(-\hat{j}) - \rho\omega\hat{j} = 0$$

$$0 = R\Omega\hat{\phi} - \rho\omega\hat{\phi} = 0$$

no slip $\qquad R\Omega - \rho\omega = 0 \qquad \omega = \frac{R}{\rho}\Omega = \frac{R}{\rho}\frac{v}{R} \qquad \omega = \frac{v}{\rho}$

$$\vec{r}_{P/O} = R\hat{r} - \rho\sin\theta\hat{r} + \rho\cos\theta\hat{z} + \rho\hat{k}$$

$$\vec{v}_{P/O} = R\dot{\hat{r}} - \rho\sin\theta\dot{\hat{r}} + \rho\cos\theta\dot{\hat{z}} + \rho\dot{\hat{k}}$$

$$= (R - \rho\sin\theta)(\Omega\hat{z}\times\hat{r}) + \rho\left[(\Omega\hat{z} - \omega\hat{i})\times\hat{k}\right]$$

$$= \Omega(R - \rho\sin\theta)\hat{\phi} + \rho\Omega(\hat{z}\times\hat{k}) - \rho\omega(\hat{i}\times\hat{k})$$

$$= \Omega(R - \rho\sin\theta)\hat{\phi} + \rho\Omega(\hat{z}\times\hat{k}) + \rho\omega\hat{j}$$

Question assigned to the following page: 2

$$\dots \Omega\left(\hat{z}\times\dot{\hat{k}}\right) + \rho\omega\left[\left(\Omega\hat{z}-\omega\hat{i}\right)\times\hat{j}\right]$$

(1) ✓ (2) (3)

(1) $= \Omega\left(R-\rho\sin\theta\right)\dot{\hat{\phi}} = \Omega\left(R-\rho\sin\theta\right)\left(\Omega\hat{z}\times\hat{\phi}\right) = \Omega^2\left(R-\rho\sin\theta\right)\left(-\hat{r}\right) = -\Omega^2\left(R-\rho\sin\theta\right)\hat{r}$

(2) $= \rho\Omega\left(\hat{z}\times\dot{\hat{i}}\right) = \rho\Omega\,\hat{z}\times\left[\left(\Omega\hat{z}-\omega\hat{i}\right)\times\hat{k}\right] = \rho\Omega\hat{z}\times\left[\Omega\hat{z}\times\hat{k} - \omega\,\hat{i}\times\hat{k}\right]$

$= \rho\Omega^2\hat{z}\times\left(\hat{z}\times\hat{k}\right) + \rho\Omega\omega\,\hat{z}\times\hat{j}$

$= \rho\Omega^2\hat{z}\times\left[\sin\theta\left(-\hat{j}\right)\right] + \rho\Omega\omega\left(-\hat{r}\right) = -\rho\Omega^2\sin\theta\,\hat{z}\times\hat{j} - \rho\Omega\omega\hat{r}$

$= \rho\Omega^2\sin\theta\hat{i} - \rho\Omega\omega\hat{r}$

(3) $= \rho\omega\left[\Omega\hat{z}\times\hat{j} - \omega\hat{i}\times\hat{j}\right] = \rho\omega\Omega\,\hat{z}\times\hat{j} - \rho\omega^2\hat{k}$

$= \rho\omega\Omega\left(-\hat{i}\right) - \rho\omega^2\left(\cos\theta\hat{z} - \sin\theta\hat{i}\right)$

$= -\rho\omega\Omega\hat{i} - \rho\omega^2\cos\theta\hat{z} + \rho\omega^2\sin\theta\hat{r}$

$$\vec{a}_{r/o} = \Omega^2\left(\rho\sin\theta - R\right)\hat{r} + \rho\Omega\left(\Omega\sin\theta - \omega\right)\hat{i} - \rho\omega\Omega\hat{i} - \rho\omega^2\cos\theta\hat{z} + \rho\omega^2\sin\theta\hat{r}$$

$$= \left[\Omega^2\rho\sin\theta - \Omega^2 R + \rho\Omega^2\sin\theta - \rho\Omega\omega - \rho\omega\Omega + \rho\omega^2\sin\theta\right]\hat{r} - \rho\omega^2\cos\theta\hat{z}$$

$$= \left[\frac{V^2}{R^2}\rho\sin\theta - \frac{V^2}{R^2} + \frac{\rho V^2}{R^2}\sin\theta - \frac{V^2}{R\rho} - \frac{V^2}{R\rho} + \frac{\rho V^2}{\rho^2}\sin\theta\right]\hat{i} - \frac{\rho V^2}{\rho^2}\cos\theta\hat{z}$$

$$= \left[\frac{V^2\rho}{R^2}\sin\theta - \frac{3V^2}{R} + \frac{\rho V^2}{R^2}\sin\theta + \frac{V^2}{\rho}\sin\theta\right]\hat{r} - \frac{V^2}{\rho}\cos\theta\hat{z}$$

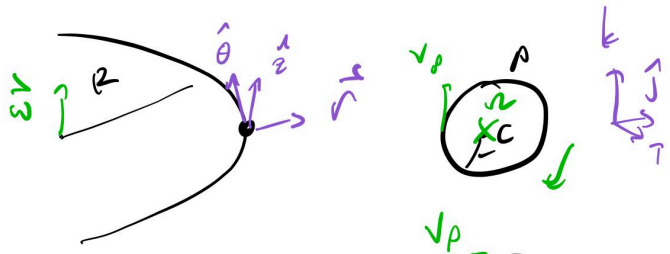
$$= \frac{\left(10\frac{m}{s}\right)(0.4m)}{30m}\sin 15° - \frac{3\left(10\frac{m}{s}\right)^2}{30m} + \frac{(0.4m)\left(10\frac{m}{s}\right)^2}{(30m)^2} + \frac{\left(10\frac{m}{s}\right)^2}{0.4m}\sin 15°\Big]\hat{r} - \frac{\left(10\frac{m}{s}\right)^2}{0.4}\cos 15°\,\hat{z}$$

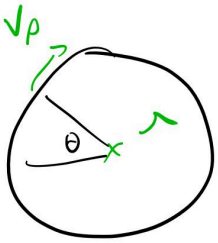
$$\vec{a}_{r/o} \simeq 54.78\,\hat{r} - 241.48\,\hat{z}$$

Question assigned to the following page: 3

...adius R.

The propeller is of radius r and rotates about its axis in a clockwise sense (as viewed from the rear) with a constant angular velocity Ω. Find the total acceleration of a point P at the tip of the propeller, assuming that its axis is always aligned with the flight path. use cylindrical unit vectors and assume that the velocity of P relative to the airplane is vertically upward at $t = 0$.





$$\hat{j} = \cos\theta\,\hat{r} - \sin\theta\,\hat{z} = \cos\Omega t\,\hat{r} - \sin\Omega t\,\hat{z}$$

$$\vec{r}\big|_P = -r\hat{j} = -r\left(\cos\Omega t\,\hat{r} - \sin\Omega t\,\hat{z}\right)$$

$$= -r\cos\Omega t\,\hat{r} + r\sin\Omega t\,\hat{z}$$

$$\dot{\hat{r}} = \omega\hat{z}\times\hat{r} = \frac{v}{R}\hat{\theta} \qquad \dot{\hat{\theta}} = \omega\hat{z}\times\hat{\theta} = -\frac{v}{R}\hat{r} \qquad \dot{\hat{z}} = \omega\hat{z}\times\hat{z} = 0$$

$$\vec{v}\big|_P = -r(-\sin\Omega t)\Omega\,\hat{r} - r\cos\Omega t\,\dot{\hat{r}} + r\cos(\Omega t)\Omega\hat{z} + r\sin(\Omega t)\dot{\hat{z}}$$

$$= r\Omega\sin\Omega t\,\hat{r} - r\cos\Omega t\left(\frac{v}{R}\hat{\theta}\right) + r\Omega\cos(\Omega t)\,\hat{z}$$

$$= r\Omega\sin\Omega t\,\hat{r} - v\frac{r}{R}\cos\Omega t\,\hat{\theta} + r\Omega\cos\Omega t\,\hat{z}$$

$$\vec{a}_P = r\Omega^2\cos\Omega t\,\hat{r} + r\Omega\sin\Omega t\,\dot{\hat{r}} + v\Omega\frac{r}{R}\sin\Omega t\,\hat{\theta} - v\frac{r}{R}\cos\Omega t\,\dot{\hat{\theta}} - r\Omega^2\sin\Omega t\,\hat{z}$$

$$= r\Omega^2\cos\Omega t\,\hat{r} + r\Omega\sin\Omega t\left(\frac{v}{R}\hat{\theta}\right) + v\Omega\frac{r}{R}\sin\Omega t\,\hat{\theta} - v\frac{r}{R}\cos\Omega t\left(-\frac{v}{R}\hat{r}\right) - r\Omega^2\sin\Omega t\,\hat{z}$$

$$= r\cos\Omega t\left(\Omega^2 + \frac{v^2}{R^2}\right)\hat{r} + \frac{2rv\Omega}{R}\sin\Omega t\,\hat{\theta} - r\Omega^2\sin\Omega t\,\hat{z}$$

$$\vec{a}_{P/O} = \frac{v^2}{R}\hat{r} \qquad \vec{a}_O = \vec{a}_P - \vec{a}_{P/O}$$

$$\boxed{\vec{a}_O = \left[-\frac{v^2}{R} + r\cos(\Omega t)\left(\Omega^2 + \frac{v^2}{R^2}\right)\right]\hat{r} + \frac{2rv\Omega}{R}\sin(\Omega t)\,\hat{\theta} - r\Omega^2\sin(\Omega t)\,\hat{z}}$$

Question assigned to the following page: 4

position shown. If the tube is frictionless, find the direction and magnitude of the velocity of the particle as it leaves the tube.

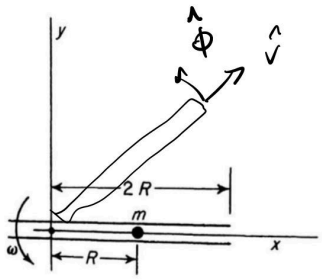


Figure P3-2

y-axis is constrained

$\Sigma F_x = ma$

$\rightarrow ma_c \quad ma_c = ma$

$\omega^2 r = \ddot{r}$

$\ddot{r} - \omega^2 r = 0$

$r(0) = R$
$\dot{r}(0) = 0$
$\ddot{r}(0) = \omega^2 R$

Assume form $r = C_1 e^{at} + C_2 e^{-at}$

$$\ddot{r} - \omega^2 r = \frac{d}{dt}\left(C_1 a e^{at} - C_2 a e^{-at}\right) - \omega^2\left(C_1 e^{at} + C_2 e^{-at}\right) = 0$$

$$C_1 a^2 e^{at} + C_2 a^2 e^{-at} - \omega^2 C_1 e^{at} - \omega^2 C_2 e^{-at} = 0$$

$\dot{r}(0) = 0 = C_1 a - C_2 a = 0 \rightarrow C_1 = C_2$

$r(0) = R = C_1 + C_2 = R \longrightarrow C_1 + (C_1) = R \quad C_1 = \frac{R}{2} \quad C_2 = \frac{R}{2}$

$$C_1 a^2 e^{at} + C_2 a^2 e^{-at} - \omega^2 C_1 e^{at} - \omega^2 C_2 e^{-at} = 0$$

at $t=0$ $\quad C_1 a^2 + C_2 a^2 - \omega^2 C_1 - \omega^2 C_2 = 0$

$$(C_1 + C_2)a^2 - \omega^2(C_1 + C_2) = 0$$

$$Ra^2 - \omega^2 R = 0 \qquad a = \omega$$

$r(t) = \frac{R}{2} e^{\omega t} + \frac{R}{2} e^{-\omega t}$

$r(t') = 2R = \frac{R}{2} e^{\omega t'} + \frac{R}{2} e^{-\omega t'} \qquad 2 = \frac{e^{\omega t'} + e^{-\omega t'}}{2} = \cosh(\omega t') \qquad t' = \frac{1}{\omega}\cosh^{-1}(2)$

$$v(t') = \dot{r}(t') = \frac{R\omega}{2} e^{\omega t} - \frac{R\omega}{2} e^{-\omega t} = R\omega \frac{e^{\omega t} - e^{-\omega t}}{2} = R\omega \sinh(\omega t')$$

$$\cosh(\omega t_f)^2 - \sinh(\omega t_f)^2 = 1 = 2^2 - \sinh(\omega t_f)^2 = 1 \rightarrow \sinh(\omega t_f) = \sqrt{3}$$

$$V_r(t_f) = R\omega \sinh(\omega t_f) = \sqrt{3} R\omega \hat{r} \qquad V_\phi = \omega(2R) = 2\omega R \hat{\phi}$$

Question assigned to the following page: 4

$\|v\| = \sqrt{v_r^2 + v_\phi^2} = \sqrt{(\sqrt{3}Rw)^2 + (2wR)^2} = wR\sqrt{7+4}$

$\boxed{\|v\| = \sqrt{7}Rw}$

$\phi_f = wt' = wt = w\left(\frac{1}{w}\cosh^{-1}(2)\right)$ $\quad \phi_f = \cosh^{-1}(2) \approx 75.46^\circ$

angle of rod w.r.t. x-axis

$\hat{r} = \cos\phi_f \hat{i} + \sin\phi_f \hat{j}$ $\quad \hat{\phi} = -\sin\phi_f \hat{i} + \cos\phi_f \hat{j}$ $\quad \phi_f =$ angle of tube at t_f

$\vec{v} = \sqrt{3}Rw\,\hat{r} + 2wR\hat{\phi} = \sqrt{3}Rw(\cos\phi_f \hat{i} + \sin\phi_f \hat{j}) + 2wR(-\sin\phi \hat{i} + \cos\phi \hat{j})$

$\vec{v}_f = Rw(\sqrt{3}\cos\phi_{f_{jp}} - 2\sin\phi_{f_{jp}})\hat{i} + Rw(2\cos\phi_{f_{jp}} + \sqrt{3}\sin\phi_{f_{jp}})\hat{j}$

$\boxed{\vec{v}_f = (-1.50\hat{i} + 2.18\hat{j})Rw}$

$\theta_f = \tan^{-1}\left(\frac{v_{f_y}}{v_{f_x}}\right) = \tan^{-1}\left(\frac{2.18}{-1.5}\right) = -55.47 + 180^\circ$

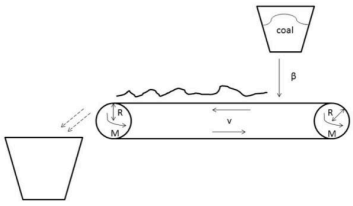
$Q_4 \to Q_2$

$\boxed{\theta_f = 124.5^\circ}$

w.r.t cartesian

Question assigned to the following page: 5

belt has identical pulleys of radius **R** at each end that rotates the belt, what is the moment **M** that should be applied to each pulley to maintain the speed of the belt constant at v?



(A poorly drawn Question 5 set-up)

$F = \frac{dp}{dt} = \cancel{m\dot{v}}^{0} + \dot{m}v$

$F = \dot{m}v = \beta v$ $\quad \Sigma M = 2M = FR = \beta vR$

$M = \frac{\beta vR}{2}$

Question assigned to the following page: 6

6) The moon sits 384,400 km away, with a satellite orbiting at radius r with an orbital plane 45° inclination (see photo). A ground station sits at $\phi = 30°$ latitude. Find the velocity and acceleration in ENU coordinates when the ground station's velocity is exactly along the moon vector, and the satellite is crossing the moon's equator going south?

ω_e = Earth's angular velocity.

